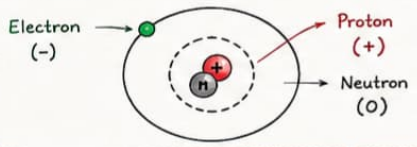


1 INTRODUCTION

- MRI uses strong magnetic field, radiofrequency (RF) pulses and gradient fields to produce images.
- Based on the behavior of MR active nuclei, mainly Hydrogen (^1H).
- No ionizing radiation $\rightarrow$ Safe.
- Excellent soft tissue contrast.
- Images depend on T_1 relaxation, T_2 relaxation and Proton density.

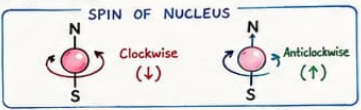
2 ATOMIC STRUCTURE

- An atom has a nucleus (protons + neutrons) surrounded by electrons in orbital shells.
- Proton $\rightarrow$ positively charged (+).
- Neutron $\rightarrow$ neutral (0).
- Electron $\rightarrow$ negatively charged (-).
- Most of the mass is in the nucleus.
- Hydrogen (^1H) $\rightarrow$ 1 proton + 1 electron.

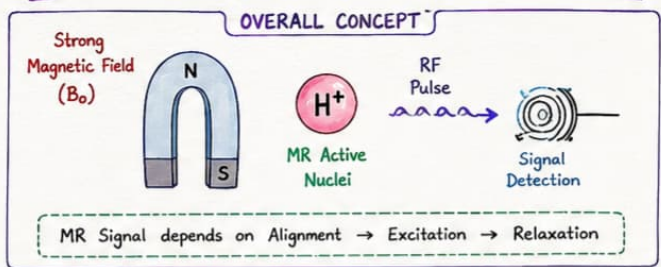


3 MOTION IN THE ATOM

- Electron $\rightarrow$ moves rapidly around the nucleus in orbital shells.
- Proton & neutron $\rightarrow$ spin on their own axis (axial spin).
- This spin creates a small magnetic field $\rightarrow$ called Spin Magnetic Moment.
- Spin can be either "clockwise" or "anticlockwise".



BASIC PRINCIPLES OF MRI



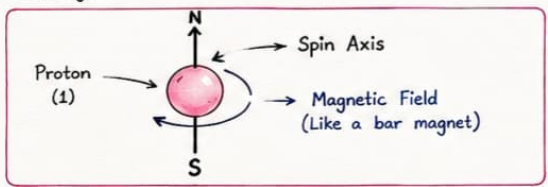
4 MR ACTIVE NUCLEI

- Nuclei having odd number of protons and/or neutrons show net spin $\rightarrow$ behave like tiny magnets.
- Only these nuclei produce MR signal.
- Common MR active nuclei:

Nucleus	Abundance in Body	Spin (I)	Relative Sensitivity
^1H (Hydrogen)	~ 63% (in water & fat)	$\frac{1}{2}$	1 (Highest)
^{13}C (Carbon)	~ 1.1%	$\frac{1}{2}$	0.015
^{15}N (Nitrogen)	Traces	$\frac{1}{2}$	0.001
^{31}P (Phosphorus)	Traces	$\frac{1}{2}$	0.06

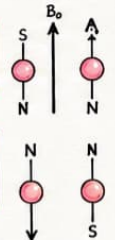
5 THE HYDROGEN NUCLEUS (^1H)

- Hydrogen has 1 proton, no neutron.
- Proton has a positive charge and spins on its axis.
- This spin creates a tiny magnetic field $\rightarrow$ acts like a bar magnet.
- ^1H is most abundant in body water & fat $\rightarrow$ main source of MR signal.



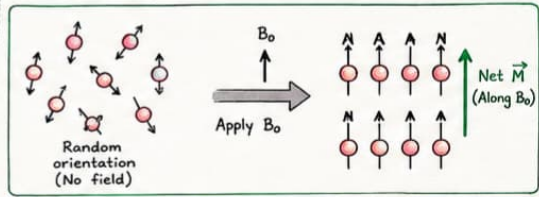
6 ALIGNMENT

- When placed in a strong magnetic field (B_0), the tiny magnets (protons) align in two ways:
 - Parallel to B_0 (Low energy)
North pole towards South pole of B_0 .
 - Anti-parallel to B_0 (High energy)
South pole towards South pole of B_0 .
- Slightly more protons align parallel (low energy) $\rightarrow$ lower energy state is preferred.



7 NET MAGNETIC VECTOR $\vec{M}$

- Because of more parallel alignment, there is a resultant magnetization along B_0 .
- This resultant is called Net Magnetic Vector ($\vec{M}$).
- $\vec{M}$ is along the axis of B_0 in the longitudinal direction (z-axis).
- No net magnetization in transverse (x-y) plane in the resting state.



HIGH YIELD EXAM FACTS

- MRI works on Nuclear Magnetic Resonance of ^1H .
- Protons act like tiny magnets due to spin.
- In B_0 field $\rightarrow$ Parallel (low energy) > Anti-parallel (high energy).
- Net magnetization ($\vec{M}$) is along B_0 (Longitudinal, z-axis).
- RF pulse tips $\vec{M}$ away from B_0 $\rightarrow$ MR signal is generated.
- Signal returns to equilibrium $\rightarrow$ Relaxation (T_1 & T_2).

1 PRECESSION AND PRECESSIONAL (LARMOR) FREQUENCY

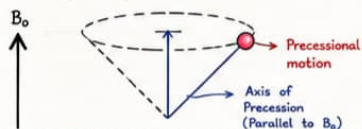
- When placed in a static magnetic field (B_0), spinning protons precess (wobble) around the direction of B_0 .
- This is like a spinning top precessing in gravity.
- The frequency of precession is called Larmor frequency (ω_0 or f_0).
- Directly proportional to magnetic field strength.

Larmor Frequency Equation

$$\omega_0 = \gamma B_0 \quad (\text{rad/s})$$

$$f_0 = \frac{\omega_0}{2\pi} = \frac{\gamma}{2\pi} B_0 \quad (\text{Hz})$$

γ = gyromagnetic ratio of proton = 42.58 MHz/T

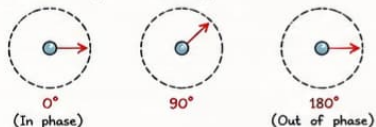


Typical Values

- At 1.5 Tesla $\rightarrow f_0 \approx 63.9$ MHz
- At 3 Tesla $\rightarrow f_0 \approx 127.7$ MHz

2 PRECESSIONAL PHASE

- Phase = position of a spin in its precessional cycle.
- Measured in degrees ($^\circ$) or radians.
- Spins at different locations get out of phase due to field inhomogeneities.
- Phase differences are the basis of MR imaging (used for spatial encoding).



Key Point

In phase spins add up $\rightarrow$ strong signal
Out of phase spins cancel out $\rightarrow$ weak/no signal

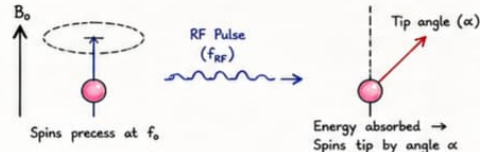
★ BASIC PRINCIPLES OF MRI ★

CONCEPT FLOW



3 RESONANCE

- Resonance occurs when the RF pulse frequency (f_{RF}) = Larmor frequency (f_0).
- Energy is absorbed efficiently $\rightarrow$ spins tip away from B_0 .
- Off-resonance $\rightarrow$ less energy absorbed $\rightarrow$ weak signal.

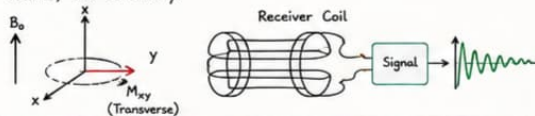


Common Tip Angles

- $90^\circ \rightarrow$ Maximum transverse magnetization (M_{xy})
- $180^\circ \rightarrow$ Inversion of longitudinal magnetization (M_z)

4 MR SIGNAL

- After excitation, transverse magnetization (M_{xy}) induces voltage in receiver coil $\rightarrow$ this is the MR signal.
- Signal comes from precessing spins in phase.
- Signal strength depends on: proton density, T_1 , T_2 , flip angle, voxel volume, coil sensitivity.

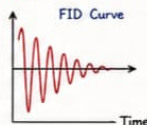


$$\text{MR Signal} \propto \text{Number of in-phase spins} \times \text{Precessional frequency}$$

More M_{xy}
(Strong signal)
More in-phase spins
High SNR

5 THE FREE INDUCTION DECAY (FID) SIGNAL

- Immediately after RF pulse off, M_{xy} starts dephasing.
- Net M_{xy} decreases $\rightarrow$ voltage in coil decays exponentially.
- This decaying signal is called Free Induction Decay (FID).
- FID contains all frequencies $\rightarrow$ basis of frequency encoding.

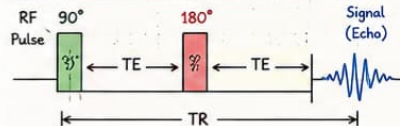


Mathematical Form

$$S(t) = S_0 e^{-t/T_2^*} \cos(\omega_0 t + \phi)$$

T_2^* = apparent transverse relaxation time

6 PULSE TIMING PARAMETERS



- TR (Repetition Time):** Time between successive RF pulses (start to start).
 - Short TR $\rightarrow$ more T_1 weighting
 - Long TR $\rightarrow$ more T_2 weighting
- TE (Echo Time):** Time between center of RF excitation and center of echo.
 - Short TE $\rightarrow$ less T_2 weighting
 - Long TE $\rightarrow$ more T_2 weighting
- Flip Angle (α):** Angle by which M is tipped by RF pulse (e.g., 90° , 180°).
- TI (Inversion Time):** Time between 180° inversion pulse and center of excitation pulse.
 - Used in Inversion Recovery sequences.
- Bandwidth (BW):** Range of frequencies acquired.
 - High BW $\rightarrow$ less noise, less SNR
 - Low BW $\rightarrow$ more noise, more SNR

ADVANTAGES

- Excellent soft tissue contrast
- Multi-planar imaging
- No ionizing radiation
- Functional & metabolic imaging
- High spatial resolution

DISADVANTAGES

- Longer scan time
- Sensitive to motion
- Expensive
- Contraindicated in some implants (pacemaker, ferromagnetic clips)

★ HIGH YIELD FACTS

- Protons behave like tiny magnets and precess in B_0 .
- f_0 (MHz) = 42.58 $\times B_0$ (Tesla)
- Resonance occurs when $f_{RF} = f_0$.
- 90° pulse $\rightarrow$ maximum M_{xy} ; 180° pulse $\rightarrow$ inversion.
- FID is the raw MR signal (source of data).
- Image contrast depends on TR, TTE, PD, T_1 , T_2 .
- SNR $\uparrow$ with field strength, voxel size, NEX, coil efficiency.

KEY TERMS: B_0 : Static Magnetic Field

TR : Repetition Time

f_0 : Larmor Frequency

TE : Echo Time

ω_0 : Angular Larmor Frequency

TI : Inversion Time

M_{xy} : Transverse Magnetization

FID : Free Induction Decay

M_z : Longitudinal Magnetization

SNR : Signal to Noise Ratio PD : Proton Density

1 INTRODUCTION

- MRI contrast is the difference in signal intensity between tissues on an image.
- It depends on tissue properties (proton density and relaxation characteristics).
- Image weighting is determined by sequence parameters that control T_1 , T_2 and PD effects.
- Goal → maximize contrast between normal and abnormal tissues.
- Good contrast → better diagnosis!

2 IMAGE CONTRAST

- Contrast arises from differences in:
 - T_1 relaxation time
 - T_2 relaxation time
 - Proton density (PD)
 - Flow, diffusion, susceptibility
- Controlled by sequence parameters:
 - TR (Repetition Time)
 - TE (Echo Time)
 - Flip angle (α)
 - Inversion time (TI)
- Different parameter settings → different image weighting.

TISSUE PROPERTIES		
		
High Water (↑T ₁ , ↑PD)	Fat (Short T ₁)	Fibrous Tissue (Long T ₁ , Low PD)

COMMON WEIGHTING & APPEARANCE

Sequence (Weighting)	TR	TE	Primary Factor	Fat	Water/CSF
T ₁ Weighted	Short	Short	T ₁	Bright	Dark
T ₂ Weighted	Long	Long	T ₂	Intermediate	Bright
PD Weighted	Long	Short	PD	Bright	Bright
STIR / FS (T ₂ W)	Long	Long	T ₂ (Fat Supp.)	Dark	Bright

(Fat suppression → removes fat signal to highlight pathology)



ADVANTAGES

- Excellent soft tissue contrast.
- Multi-contrast without radiation.
- Functional information (DWI, PWI, MRS).
- Image weighting can be tailored.

KEY TERMS:

TR : Repetition Time
T₁ : Spin-Lattice Relaxation Time (Longitudinal)



DISADVANTAGES

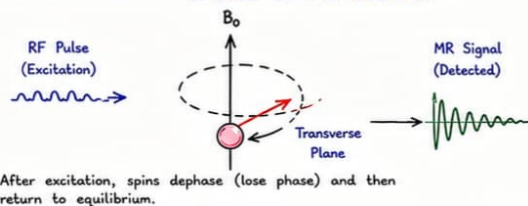
- Longer scan time.
- Sensitive to patient motion.
- Susceptibility artifacts.
- Contraindications (implants, pacemakers).

TE : Echo Time
T₂ : Spin-Spin Relaxation Time (Transverse)

TI : Inversion Time
T₂ : Spin-Spin Relaxation Time (Transverse)

★ IMAGE WEIGHING AND CONTRAST ★

BASICS OF MR SIGNAL



3 RELAXATION

- After excitation, spins return to equilibrium by two processes:
 - T₁ Relaxation → recovery of M_z along B₀ (spin-lattice interaction).
 - T₂ Relaxation → decay of M_{xy} in transverse plane (spin-spin interaction).
- These determine the MR signal and image contrast.

RELAXATION SUMMARY

T ₁ (Longitudinal)	T ₂ (Transverse)
Recovery of M _z (along B ₀)	Decay of M _{xy} (in transverse plane)
Spin-lattice relaxation	Spin-spin relaxation
Tissue specific time	Tissue specific time
Controls T ₁ weighting	Controls T ₂ weighting

RELAXATION HIERARCHY

Short → Long				
T ₁	Fat	Liver	Muscle	CSF
T ₂	Fat	Muscle	Liver	CSF

PROTON DENSITY (PD)

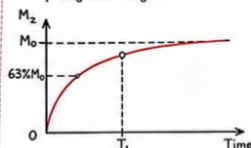
- Refers to number of hydrogen protons in a given volume of tissue.
- Higher PD → higher signal on PD weighted images.

PD Order (High → Low)

CSF > Edema > Muscle > Fat > Bone

4 T1 RECOVERY (LONGITUDINAL)

- Recovery of longitudinal magnetization (M_z) along B₀ after RF pulse.
- Described by T₁ (spin-lattice relaxation time).
- Short T₁ → fast recovery → more signal on T₁ weighted images.
- Long T₁ → slow recovery → low signal on T₁ weighted images.

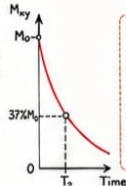


Key Points

- At time = T₁, M_z recovers to 63% of M₀.
- ~5T₁ → 99% recovery.

5 T2 DECAY (TRANSVERSE)

- Decay of transverse magnetization (M_{xy}) due to loss of phase coherence.
- Described by T₂ (spin-spin relaxation time).
- Long T₂ → slow decay → more signal on T₂ weighted images.
- Short T₂ → fast decay → low signal on T₂ weighted images.



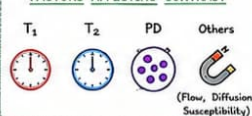
Key Points

- At time = T₂, M_{xy} decays to 37% of M₀.
- ~5T₂ → ~0 (no transverse signal).

6 CONTRAST MECHANISM

- Contrast depends on relative contribution T₁, T₂ and PD in a sequence.
- General rules:
 - Short TR → T₁ weighting
 - Long TR → T₂ or PD weighting
 - Short TE → T₁ or PD weighting
 - Long TE → T₂ weighting
 - Fat suppression → improves lesion detection.

FACTORS AFFECTING CONTRAST



HIGH YIELD EXAM FACTS

- T₁ short tissues → Fat < Liver < Muscle < CSF (longest)
- T₂ long tissues → Fat < Muscle < Liver < CSF (longest)
- At time = T₁ → M_z = 63% of M₀
- At time = T₂ → M_{xy} = 37% of M₀

- T₁ weighting → Short TR, Short TE
- T₂ weighting → Long TR, Long TE
- PD weighting → Long TR, Short TE
- Fat suppression important in T₂ imaging

α : Flip Angle
M_z : Longitudinal Magnetization

PD : Proton Density
M_{xy} : Transverse Magnetization



MRI - IMAGE WEIGHTING & CONTRAST



① RELAXATION IN DIFFERENT TISSUES

- MRI signal depends on how tissues relax after RF pulse → **T₁** (longitudinal) & **T₂** (transverse)

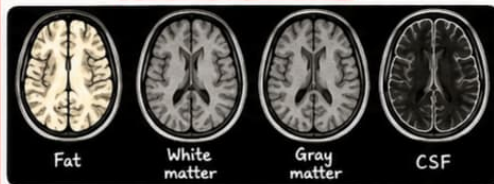
TISSUE	T ₁ RELAXATION	T ₂ RELAXATION
Fat	Very Short	Short
White matter	Short	Short
Gray matter	Intermediate	Intermediate
CSF / Water	Long	Long
Calcification / Cortical bone	Very Short	Very Short
Air	No signal	No signal

- T₁** : recovery of longitudinal magnetization (M_z)
- T₂** : decay of transverse magnetization (M_{xy})

② T₁ CONTRAST

- Based on differences in T₁ relaxation times
- TR short, TE short**
- Tissues with **short T₁** → appear **bright**

T₁ WEIGHTED IMAGE

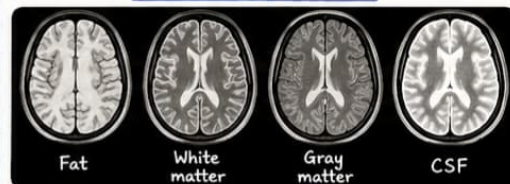


Brightness : Fat > White matter > Gray matter > CSF

③ T₂ CONTRAST

- Based on differences in T₂ relaxation times
- TR long, TE long**
- Tissues with **long T₂** → appear **bright**

T₂ WEIGHTED IMAGE



Brightness : CSF > Gray matter > White matter > Fat

④ PROTON DENSITY CONTRAST

- Based on number of hydrogen protons (density)
- TR long, TE short**
- Reflects relative proton concentration
- Less T₁ or T₂ influence

PD WEIGHTED IMAGE



Brightness : Depends on proton density
(Fat ≈ White matter > Gray matter > CSF)

⑤ WEIGHTING - HOW TO ACHIEVE ?

WEIGHTING	TR	TE	MAIN FACTOR	APPEARANCE (BRIGHT)
T ₁ weighted	Short	Short	T ₁ differences	Fat, WM > GM > CSF
T ₂ weighted	Long	Long	T ₂ differences	CSF > GM > WM > Fat
PD weighted	Long	Short	Proton density	Fat ≈ WM > GM > CSF
STIR / FLAIR (Inversion recovery)	Long	Long	T ₁ suppressed	Fluid/edema bright, Fat dark

TR : Repetition Time **TE** : Echo Time
Short TR → less time for recovery (T₁ effect ↑)
Long TE → more dephasing (T₂ effect ↑)

⑥ OTHER CONTRAST MECHANISMS

- Chemical Shift** : Fat & water resonate at slightly different frequencies → in-phase / out-of-phase
- Flow related** : Time-of-flight, Phase contrast (blood appears bright or dark based on sequence)
- Diffusion** : DWI / ADC → detects Brownian motion (restricted diffusion appears bright on DWI)
- Paramagnetic contrast agents (Gd based)** : Shorten T₁ → enhance tissues
- Susceptibility effects** : Hemorrhage, calcification, air → signal loss (blooming)

✓ KEY POINTS

- T₁ short tissues → Bright on T₁WI (Fat, WM)
- T₂ long tissues → Bright on T₂WI (CSF, edema)
- PD reflects proton concentration (less T₁/T₂ effect)
- Image contrast = TR, TE, & tissue properties
- Know the **BRIGHT** and **DARK** for exam!

★ HIGH-YIELD EXAM FACTS

- Fat is bright on T₁WI, dark on STIR.
- CSF is dark on T₁WI, bright on T₂WI & FLAIR.
- Gray matter is always brighter than white matter on T₂WI.
- Short TR + Short TE = T₁ weighting.
- Long TR + Long TE = T₂ weighting.
- PDW uses Long TR + Short TE.
- Gadolinium → Shortens T₁ → enhancement on T₁WI.



⚠ ADVANTAGES & LIMITATIONS

ADVANTAGES

- Excellent soft tissue contrast
- Multiplanar imaging
- No ionizing radiation
- Functional information (diffusion, perfusion, flow)

LIMITATIONS

- Longer scan time
- Motion sensitive
- Expensive
- Contraindicated in some implants, pacemakers

REMEMBER : " **TR** decides T₁ effect, **TE** decides T₂ effect, Tissue decides the signal! "



★ SPIN ECHO PULSE SEQUENCE ★

① INTRODUCTION

- Most widely used MRI pulse sequence.
- Produces high-quality T1, T2 & PD weighted images.
- Uses 90° excitation pulse followed by 180° refocusing pulse to form an echo.
- Echo forms at time TE (spin echo).
- Less sensitive to B₀ inhomogeneity than gradient echo.
- Basis for many 2D & 3D sequences.

IMPORTANT COMPONENTS

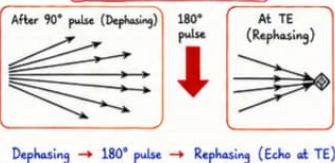
- 90° RF pulse (excitation)
- 180° RF pulse (refocusing)
- Slice selection gradients (G_{ss})
- Phase encoding gradient (G_{pe})
- Readout gradient (G_{ro})
- ADC (data acquisition)
- TR, TE, Flip angles



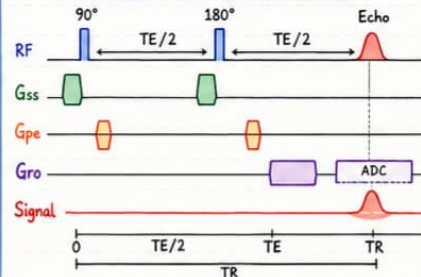
② RF REPHASING (PRINCIPLE)

- After 90° pulse, spins dephase due to:
 - Differences in Larmor frequency (T2*)
 - Magnetic field inhomogeneity (ΔB_0)
- 180° pulse inverts spins.
- Faster spins slow down & slower spins catch up.
- At TE, all spins are in phase → coherent signal (echo).
- Not true refocusing of T2 decay, but rephasing of dephased spins.

CONCEPT OF REPHASING



③ CONVENTIONAL SPIN ECHO

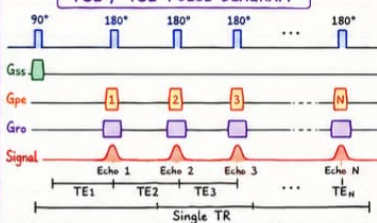


- One echo per TR.
- Echo at TE determines image contrast.
- TR controls T1 weighting; TE controls T2 weighting.
- Long acquisition time for multiple slices.

④ FAST OR TURBO SPIN ECHO (FSE / TSE)

- Multiple echoes generated after a single 90° excitation.
- 180° pulses applied in a train with varying phase encoding.
- Greatly reduces scan time.
- Commonly used for T2-weighted imaging.

FSE / TSE PULSE DIAGRAM



- Echo train length (ETL) = number of echoes.
- Longer ETL → faster scan, but ↑ T2 blurring.

COMPARISON: CONVENTIONAL vs FSE/TSE

Feature	Conventional Spin Echo	FSE / TSE
Echoes per TR	1	Multiple
Scan Time	Long	Shorter (faster)
SNR	Higher	Slightly lower
T2 Blurring	Minimal	Increases with ETL
SAR	Lower	Higher
Use	High resolution, small FOV, research	Routine T2WI, clinical practice

CLINICAL APPLICATIONS



- T2-weighted imaging: Brain, spine, joints, abdomen, pelvis.
- Fluid, edema, inflammation → bright on T2WI.
- FSE/TSE: Routine musculoskeletal (MSK) imaging of spine, knee, shoulder.
- Black-blood variants (e.g., STIR, TIRM) for MSK, spine.
- Heavily T2-weighted sequences (long TE) e.g., MRCP, cystography.

✓ ADVANTAGES

- Excellent T2 contrast.
- Less sensitive to B₀ inhomogeneity.
- Robust and widely available.
- FSE/TSE → faster acquisition.
- High SNR with long TR.

⚠ LIMITATIONS

- Longer scan time in conventional SE.
- T2 blurring with long ETL in FSE/TSE.
- Higher SAR with FSE/TSE.
- Lower SNR per unit time vs GRE.

★ KEY POINTS

- 90° pulse excites → 180° pulse refocuses → echo at TE.
- Echo time (TE) determines image contrast.
 - Short TE → T1/PD weighting
 - Long TE → T2 weighting
- TR controls T1 weighting.
- FSE/TSE uses echo train → faster scans.
- Echo spacing (ESP) = time between successive echoes.
- Know ETL, TE, TR, ESP for exam!

★ HIGH-YIELD EXAM FACTS

- Spin echo reduces effects of B₀ inhomogeneity.
- SE signal decays with T2 (not T2*).
- TE = time from 90° pulse to echo peak.
- Minimum TE = TE_{min} ≈ TE/2 (practical).
- FSE/TSE: Echo spacing (ESP) = echo train spacing.
- Increasing ETL → ↑ speed, ↓ SNR, ↑ T2 blurring, ↑ SAR.



SIGNAL INTENSITY (RELATIVE)

Tissue	T1WI (Short TE/Short TR)	T2WI (Long TE/Long TR)
Fat	Bright	Bright
Water / CSF	Dark	Very Bright
White matter	Intermediate	Intermediate
Gray matter	Intermediate	Bright
Bone / Air	Dark (no signal)	Dark (no signal)

TIPS TO REMEMBER

- "90° excites, 180° refocuses, TE produces echo."
- "TR for T1, TE for T2."
- "Single echo per TR = Conventional SE."
- "Multiple echoes in one TR = FSE/TSE."
- "More ETL = More speed, more blur, more heat."

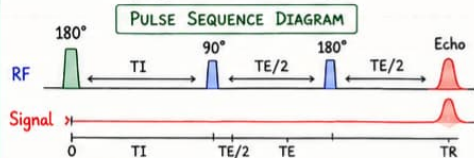


REMEMBER: "90° excites - 180° refocuses - TE makes the echo - TR resets the system!"

★ SPIN ECHO PULSE SEQUENCE - INVERSION RECOVERY (IR) FAMILY ★

① INVERSION RECOVERY (IR)

- General IR is a spin echo sequence that uses 180° inversion pulse followed by a delay (TI) before 90° excitation.
- Allows selective nulling/attenuation of tissues based on their T1 relaxation.
- Used to highlight tissues with long T1.



KEY POINTS

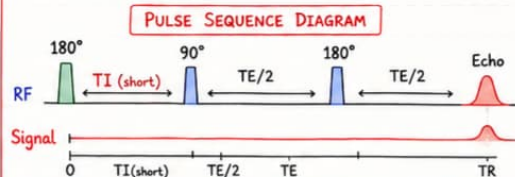
- 180° inversion $\rightarrow$ flips M_z to negative.
- During TI, M_z recovers toward $+z$.
- 90° pulse at TI time excites residual M_z .
- If $M_z = 0$ at TI $\rightarrow$ signal is nulled.

NULL CONDITION

$$TI_{null} = T1 \times \ln(2) \approx 0.693 \times T1$$

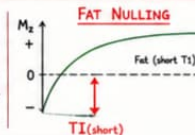
② SHORT TAU INVERSION RECOVERY (STIR)

- An IR sequence with very short TI designed to null fat.
- Fat has short T1 $\rightarrow$ recovers to zero quickly.
- Produces uniform fat suppression independent of B_0 inhomogeneity.



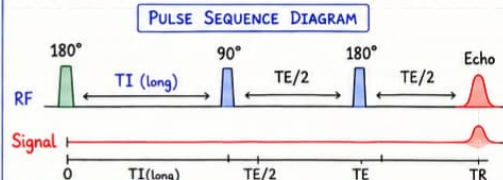
KEY POINTS

- Very short TI (typically 80-180 ms at 1.5T).
- Fat magnetization crosses zero at this TI $\rightarrow$ nulled.
- Good fat suppression even with field inhomogeneity.
- CSF and fluids remain bright (long T1).



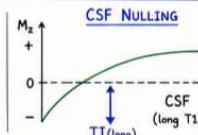
③ FLUID ATTENUATED INVERSION RECOVERY (FLAIR)

- An IR sequence with TI chosen to null CSF (long T1 fluid).
- Suppresses free fluid signal $\rightarrow$ better lesion conspicuity.
- Insensitive to flow; unlike SE, it suppresses both static and slow-flowing CSF.



KEY POINTS

- Long TI ($\approx 2000-2500$ ms at 1.5T).
- CSF magnetization crosses zero $\rightarrow$ nulled.
- White matter appears darker.
- Useful for detecting periventricular lesions.



CLINICAL APPLICATIONS

IR (General)

- Demyelinating lesions
- Tumor characterization (e.g., post-contrast)
- Spinal cord lesions
- Delayed enhancement (e.g., myocardium)

STIR

- Musculoskeletal imaging
- Bone marrow edema, fractures
- Soft tissue inflammation, myositis
- Lymphoma, metastasis (fat suppressed)

FLAIR

- Multiple sclerosis (plaques)
- Gliomas
- Subarachnoid hemorrhage
- Encephalitis, leukoencephalopathy

COMPARISON : IR vs STIR vs FLAIR

Feature	IR (General)	STIR	FLAIR
Purpose	T1 weighting / tissue selection	Fat suppression	CSF suppression
TI	Variable	Short (80-180 ms)	Long (2000-2500 ms)
Null Tissue	Depends on TI	Fat	CSF
Null Tissue Dependence	Depends on TI	Less sensitive to B_0	Less sensitive to B_0
Typical Use	General purpose	MSK, whole body	Brain, spine
Contrast	T1 weighted	T2 weighted (fat suppressed)	T2 weighted (CSF suppressed)



ADVANTAGES

- Excellent tissue contrast control.
- IR allows selective nulling of tissues.
- STIR $\rightarrow$ homogeneous fat suppression.
- FLAIR $\rightarrow$ excellent lesion detection.
- Reduces CSF flow artifacts (FLAIR).
- Widely used in neuro and MSK imaging.



LIMITATIONS

- Longer scan time (due to TI).
- Lower SNR (signal loss due to 180° inversion pulse).
- Sensitive to T1 variations (IR).
- STIR may show longer scan time than fat sat SE.
- FLAIR: incomplete CSF suppression if TI not optimized.

★ KEY POINTS

- All use 180° inversion pulse followed by TI $\rightarrow$ 90° excitation.
- TI decides which tissue is nulled based on its T1.
- Null point when $M_z = 0$ (crosses zero).
- Short TI $\rightarrow$ null short T1 (fat). Long TI $\rightarrow$ null long T1 (CSF).
- TE same as spin echo; image contrast mainly by TI.



HIGH-YIELD EXAM FACTS

- $TI_{null} = 0.693 \times T1$ (approx.).
- At 1.5T: T1 fat = 250 ms $\rightarrow$ TI(STIR) $\approx 100-150$ ms.
- T1 CSF $\approx 2000-4000$ ms $\rightarrow$ TI(FLAIR) $\approx 2000-2500$ ms.
- STIR = Short T1 Inversion Recovery = Fat null.
- FLAIR = Fluid Attenuated IR = CSF null.
- FLAIR superior to T2WI for periventricular lesions.

TIPS TO REMEMBER

- "Short TI $\rightarrow$ Fat Bye (STIR)"
- "Long TI $\rightarrow$ Fluid Bye (FLAIR)"
- "IR is the master sequence for tissue selection."
- "Null the unwanted $\rightarrow$ see the desired."



REMEMBER : IR = Selective inversion - STIR = Short TI kills fat - FLAIR = Long TI silences fluid



GRADIENT ECHO PULSE SEQUENCE



1 INTRODUCTION

- Gradient echo (GRE) is an MRI pulse sequence that uses gradient reversal instead of 180° RF pulse.
- Produces T1-weighted images (short TR, small flip angle).
- Fast acquisition → useful in dynamic, 3D and functional imaging.
- No 180° pulse → less SAR, shorter scan time, sensitive to B₀ inhomogeneity.

KEY COMPONENTS

- RF excitation pulse
- Gradient pulses (slice select, phase encode, frequency encode)
- Spoiler gradients (dephasing)
- ADC for signal acquisition



2 VARIABLE FLIP ANGLE

- Flip angle (α) controls amount of longitudinal magnetization used.
- Small α → faster recovery → more T1 weighting.
- Large α → higher signal but more saturation.
- Steady-state achieved after few TRs.
- Ernst angle (α_E) gives maximum signal for a given T1 and TR.

Ernst Angle Formula

$$\alpha_E = \cos^{-1}(e^{-TR/T1})$$

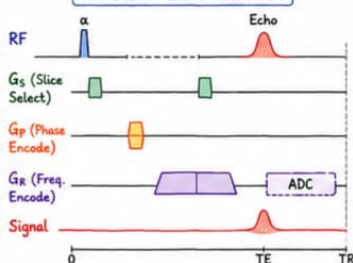
EFFECT OF FLIP ANGLE (α)

Flip Angle	Signal	T1 Weighting	SNR	SAR
Low (5°-20°)	Lower	High	Lower	Low
Medium (20°-60°)	Moderate	Moderate	Moderate	Low
High (60°-90°)	Higher	Lower	Higher	Higher

3 GRADIENT REPHASING (PRINCIPLE)

- After excitation, spins dephase due to field inhomogeneity.
- Applying a gradient with opposite polarity reverses the dephasing.
- Spins rephase → form gradient echo at TE.
- No 180° pulse is used.

GRADIENT ECHO DIAGRAM



4 WEIGHTING IN GRE

- GRE primarily gives T1 weighting.
- Depends on: TR, TE, flip angle (α).
- Short TR, short TE, small α → strong T1 weighting.
- Longer TR → more T2* influence.
- T2* decay includes true T2 + susceptibility effects.

WEIGHTING SUMMARY

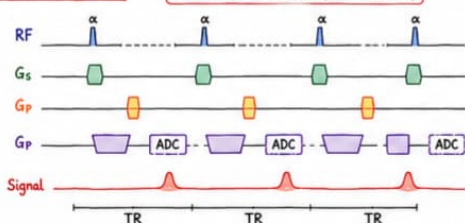
Parameter	Increase → Effect	Decrease → Effect
TR	↑ T1 weighting	↑ T2* weighting
TE	↑ T2* decay	↓ T2* decay
Flip angle (α)	↑ Signal (up to Ernst angle)	↓ Signal
Bandwidth	↓ T2* artifact	↑ Noise (↓ SNR)

T2* = Apparent T2 (includes field inhomogeneity)

5 COHERENT OR REWOUND GRADIENT ECHO

- Special GRE where gradients are balanced (rewound) → retain phase coherence.
- Maintains steady-state transverse magnetization.
- Used in steady-state or balanced GRE (SSFP) sequences.
- Produces high SNR and excellent contrast between fluids and tissues.
- Examples: Balanced SSFP, FIESTA, TrueFISP.

COHERENT GRE DIAGRAM (Balanced SSFP)



KEY FEATURES

- Gradients are fully balanced in all three axes.
- No net phase accrual between TRs.
- Steady-state signal is maintained.
- Very high SNR.

COMPARISON : SPIN ECHO vs GRADIENT ECHO

Feature	Spin Echo (SE)	Gradient Echo (GRE)
Refocusing	180° RF pulse	Gradient reversal
Sensitivity to B ₀ inhomogeneity	Less	High (causes signal loss)
T2 vs T2*	T2	T2* (T2 + susceptibility)
TR / TE	Longer / moderate	Short / short
SAR	Higher (due to 180°)	Lower
Acquisition time	Longer	Shorter (faster)
Typical use	T2-weighted imaging	T1-weighted, 3D, dynamic, functional, angiography

KEY POINTS

- GRE uses gradient reversal instead of 180° pulse.
- Fast, efficient, and sensitive to B₀ inhomogeneity.
- T1 weighting depends on TR, TE, and flip angle.
- GRE signal decays with T2*.
- Coherent GRE (balanced) → very high SNR.



ADVANTAGES

- Fast acquisition.
- No 180° pulse → lower SAR.
- Good for 3D, dynamic, perfusion, angiography, fMRI.
- High SNR in balanced GRE.

LIMITATIONS

- Sensitive to magnetic field inhomogeneity.
- Susceptibility artifacts (T2* decay).
- Lower contrast for T2-weighted imaging compared to SE.
- Flow and motion may cause artifacts.

HIGH-YIELD EXAM FACTS

- Ernst angle: $\alpha_E = \cos^{-1}(e^{-TR/T1})$
- GRE signal $\propto \sin(\alpha) (1 - e^{-TR/T1}) / (1 - \cos(\alpha)e^{-TR/T1})$
- T2* < T2 < T1
- GRE → T1 weighted | SE → T2 weighted
- Used in MRA (TOF), DCE-MRI, fMRI, VIBE, FLASH.



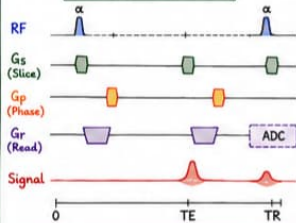
REMEMBER : GRE = No 180° - Use Gradients - Fast - T1 Weighted - Sensitive to B₀ - T2* Decay

★ GRADIENT ECHO PULSE SEQUENCES ★

① INCOHERENT OR SPOILED GRE (SPGR)

- Most common GRE sequence.
- Steady-state magnetization is spoiled between TRs to eliminate residual transverse magnetization.
- Spoiling done by:
 - RF spoiling (incremental phase)
 - Gradient spoiling (moment variation)
- Short TR, short TE, small flip angle (α).
- Produces T1-weighted images.
- Examples: SPGR, FLASH, RF-SPGR.

SPOILED GRE DIAGRAM



KEY POINTS

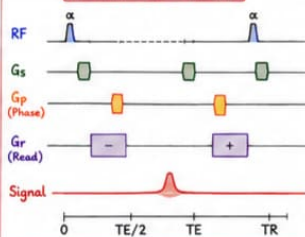
- Residual transverse magnetization is destroyed (spoiled).
- High SNR & good T1 contrast.
- Fast 3D imaging possible.



② REVERSE ECHO GRADIENT ECHO (RGE)

- Uses a 180° refocusing gradient instead of RF 180° pulse.
- Reads out the "reversed" echo formed in the middle of the read gradient.
- Less sensitive to B₀ inhomogeneity than GRE.
- Longer TE possible without extra RF pulse.
- Lower SAR compared to spin echo.

REVERSE ECHO DIAGRAM



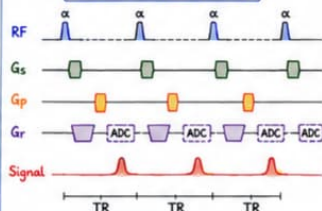
KEY POINTS

- Refocusing done by gradient reversal.
- Echo time = center of reversed read gradient.
- Less B₀ sensitivity, low SAR.
- Used in MRA (TOF, CE-MRA).

③ BALANCED GRADIENT ECHO (bGRE/SSFP)

- Gradients are balanced in all three axes within each TR.
- Net gradient moment = 0.
- Coherent steady-state achieved → high signal from T2/T1 ratio.
- Very high SNR and efficient.
- Tr/TE short; flip angle moderate (α 50°-70°).

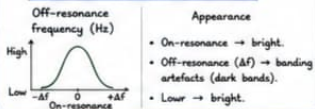
BALANCED GRE DIAGRAM



KEY POINTS

- All gradients balanced (no net moment).
- Produces bright fluid/blood (T2/T1 weighting).
- Sensitive to off-resonance & flow.
- Examples: TrueFISP, FIESTA, bFFE.

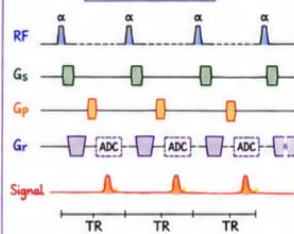
SIGNAL CHARACTERISTICS (bGRE)



④ FAST GRADIENT ECHO (FLASH)

- A type of spoiled GRE.
- Very short TR/TE.
- Very small flip angle (α 5°-30°).
- Extreme T1 weighting.
- Used for very fast 2D/3D imaging.

FLASH DIAGRAM



KEY POINTS

- Uses very short TR/TE.
- High speed, low SAR.
- Good T1 contrast.
- Examples: FLASH, 3D-FLASH.

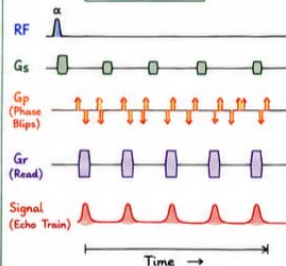
COMMON APPLICATIONS

- DCE-MRI (dynamic studies)
- 3D T1-weighted imaging
- Angiography (MRA)
- Fast whole-body imaging

⑤ ECHO PLANAR IMAGING (EPI)

- Single RF excitation (α) followed by a train of rapid gradient echoes.
- Uses "blipped" phase encoding to fill k-space line by line.
- Extremely fast (ms-sec).
- Very sensitive to susceptibility & B₀ inhomogeneity.
- Low spatial distortion correction techniques used.

EPI DIAGRAM



KEY POINTS

- Fastest MRI sequence.
- T2* weighted.
- Susceptibility artefacts common.
- Essential for fMRI & DWI.



COMPARISON OF GRE VARIANTS

Sequence	Spoiled?	Gradient Balanced?	Refocusing Mechanism	Main Weighting	SNR	Speed	B ₀ Sensitivity
Spoiled GRE (SPGR)	Yes	No	None (spoiled)	T1	High	Fast	Low-Mod
Reverse Echo GRE	No	No	180° Gradient	T1 / T2*	High	Fast	Low
Balanced GRE (bGRE)	No	Yes	Coherent SSFP	T2/T1	Very High	Fast	High
Fast GRE (FLASH)	Yes	No	None (spoiled)	T1 (extreme)	High	Very Fast	Low-Mod
EPI	Yes	No	Echo Train	T2*	Low-Mod	Extremely Fast	Very High

✓ ADVANTAGES

- Very fast imaging.
- Short scan time → patient comfort.
- High SNR in balanced GRE.
- Versatile: 2D, 3D, dynamic, MRA, fMRI.
- Lower SAR (except bGRE slightly higher).

⚠ LIMITATIONS

- Sensitive to B₀ inhomogeneity (except RGE).
- T2* decay causes signal loss.
- Susceptibility artefacts (EPI, bGRE).
- Lower tissue contrast than spin echo.
- Flow & motion artefacts.

★ HIGH-YIELD EXAM FACTS

- GRE uses gradient reversal instead of 180° pulse.
- Signal $\propto \sin(\alpha) (1 - e^{-TR/T1}) e^{-TE/T2^*}$
- Spoiled GRE → most common T1-weighted 3D.
- bGRE → very high SNR, sensitive to off-resonance.
- EPI → fastest, used in fMRI & DWI.



REMEMBER : Spoil the past (SPGR) - Reverse the gradient (RGE) - Balance the gradients (bGRE) - Fast & small flip (FLASH) - Echo train for speed (EPI)



SPATIAL ENCODING



① INTRODUCTION

- Spatial encoding localizes the MR signal from a specific position (voxel) in space.
- Uses magnetic field gradients applied in x, y and z directions.
- Together with RF excitation and signal detection → forms the basis of MRI.
- Two main encoding steps:
 - Slice selection (1D)
 - Phase & Frequency encoding (2D)
 → gives a 3D volume.

KEY CONCEPT



Position (r) is mapped to frequency (f) or phase (Φ) using gradients.

$$f = \frac{1}{2\pi} G \cdot r \quad (\text{Frequency encoding})$$

$$\Phi = \gamma G \cdot r \cdot \Delta t \quad (\text{Phase encoding})$$

γ = Gyromagnetic ratio (42.58 MHz/T for ^1H)

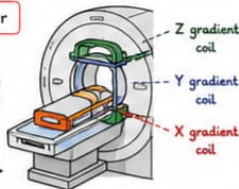
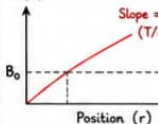
G = Gradient strength (T/m), r = position (m)

② MECHANISM OF GRADIENTS

- Gradients are linear changes in magnetic field strength along a direction.
- Created by gradient coils (X, Y, Z) placed inside the main magnet.
- Magnetic field with gradient:

$$B(r) = B_0 + G \cdot r$$

$B(T)$



- Results in position-dependent Larmor frequency:

$$\omega(r) = \gamma B(r) = \gamma (B_0 + G \cdot r)$$

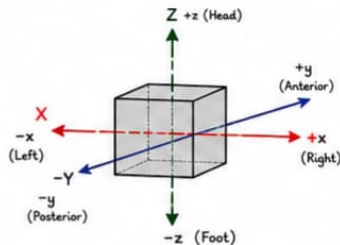
- Different positions precess at different frequencies → enables spatial encoding.

GRADIENT CHARACTERISTICS

- Rise time (T_r): time to reach target strength.
- Max amplitude ($G_{\max}$): determines resolution.
- Slew rate (S): change in gradient per ms (T/m/ms), higher S → faster imaging.

③ GRADIENT AXES

- Three mutually perpendicular gradient systems: G_x (left-right), G_y (anterior-posterior), G_z (head-foot).
- Each produces a linear field change along its axis.



$G_x \rightarrow$ Left ($-x$) to Right ($+x$)

$G_y \rightarrow$ Posterior ($-y$) to Anterior ($+y$)

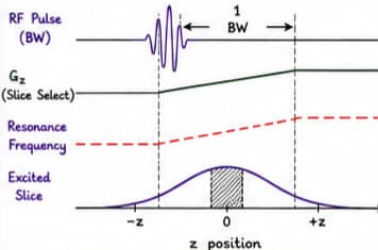
$G_z \rightarrow$ Foot ($-z$) to Head ($+z$)

- Any direction in space can be obtained by combining these three gradients.

④ SLICE SELECTION (G_z)

- Used to excite a thin slice at a desired position along the z-axis.
- Achieved using:
 - RF pulse with bandwidth (BW)
 - Slice selection gradient (G_z)
- Only spins within the BW are in resonance → excited.

SLICE SELECTION DIAGRAM



SLICE THICKNESS

$$\Delta z = \frac{BW}{\gamma G_z}$$

- Increase BW → thicker slice
- Increase G_z → thinner slice

★ KEY POINTS

- Spatial encoding = mapping position → frequency/phase.
- Gradients provide position-dependent magnetic field.
- Three linear gradients: G_x , G_y , G_z .
- Slice selection uses G_z + RF bandwidth.
- Phase encoding determines y-position (phase shift).
- Frequency encoding determines x-position (frequency of echo).



SUMMARY OF SPATIAL ENCODING

Step	Gradient	Purpose	Encoding Type
1. Slice Selection	G_z + RF pulse	Select a thin slice along z-axis	1D (along z)
2. Phase Encoding	G_y (blip)	Impose phase shift along y-axis	2D (phase)
3. Frequency Encoding	G_x (readout)	Detect signal with frequency variation along x-axis	2D (frequency)
Repetition	Repeat for all phase encodes and slices	Builds 3D volume	

✓ ADVANTAGES

- Precise localization of MR signal.
- High spatial resolution achievable.
- Flexible control of slice thickness, FOV and resolution.
- Enables fast imaging using echo-planar/parallel techniques.

⚠ LIMITATIONS

- Stronger gradients → more artifacts.
- Limited by slew rate and heating.
- Susceptible to motion during encoding.

★ HIGH-YIELD EXAM FACTS

- Gradient strength unit: Tesla per meter (T/m).
- Slew rate unit: T/m/ms.
- Gyromagnetic ratio of ^1H : 42.58 MHz/T.
- Slice thickness: $\Delta z = BW / (\gamma G_z)$.
- Phase encoding blip → no net area (rewind).
- Frequency encoding gradient → readout gradient.
- Spatial encoding converts 3D object → 3D data matrix → image.



REMEMBER : SELECT the slice (G_z) - ENCODE the phase (G_y) - READ the frequency (G_x) - REPEAT to build the image.

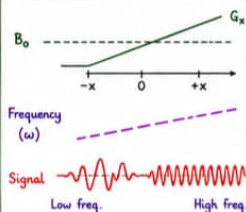
★ SPATIAL ENCODING ★

① FREQUENCY ENCODING (READOUT)

- A gradient (G_x) is applied during signal acquisition.
- Creates a linear change in magnetic field along x-axis.
- Spins at different x positions precess at different frequencies.
- Signal received is a mix of all frequencies.
- FFT converts the frequency information into spatial (x) position → forms a line of k-space.



FREQUENCY ENCODING DIAGRAM



KEY POINTS

- G_x on during readout.
- Maps position → frequency.
- One readout = one line of k-space.
- Determines x-position (frequency encoding).

EQUATION

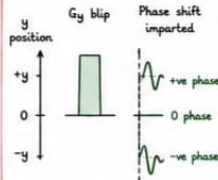
$$\omega(x) = \gamma (B_0 + G_x \cdot x)$$

$$f(x) = \frac{\omega(x)}{2\pi}$$

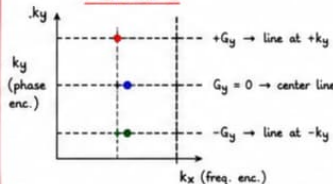
② PHASE ENCODING

- A gradient (G_y) is applied for a short time before readout.
- Imports a position-dependent phase shift along y-axis.
- No gradient during readout in y → phase differences are retained.
- Each step of G_y changes the phase → selects a different line in k-space.
- Repeated with different G_y amplitudes to fill k-space (line by line).

PHASE ENCODING DIAGRAM



k-SPACE VIEW



KEY POINTS

- G_y blip before readout.
- Controls phase of spins.
- Determines y-position (phase encoding).
- Different G_y steps → fill k-space vertically.

PHASE SHIFT EQUATION

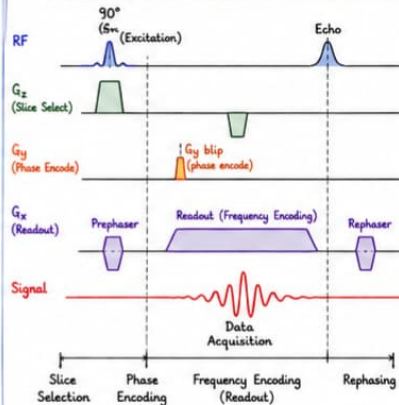
$$\phi(y) = \gamma \cdot G_y \cdot y \cdot \Delta t$$

where Δt = duration of G_y blip

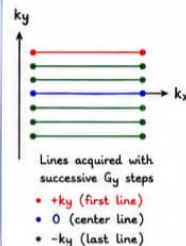
③ BRINGING IT ALL TOGETHER - PULSE SEQUENCE TIMING

- Slice selection (G_z) → excites a thin slice.
- Phase encoding (G_y blip) → selects a line (ky).
- Frequency encoding (G_x) → reads out the line while signal is acquired.
- Repeat for multiple G_y steps → fill k-space.
- Apply IFFT to k-space → final 2D image.

PULSE SEQUENCE TIMING DIAGRAM (2D GRE EXAMPLE)



k-SPACE FILLING



KEY POINTS

- G_z → slice selection (1D).
- G_y → phase encoding (2D).
- G_x → frequency encoding (2D).
- k-space filled line by line.
- k-space → image by IFFT.

★ KEY POINTS (AT A GLANCE)

- Spatial encoding = mapping position → frequency/phase.
- Three gradients: G_x , G_y , G_z .
- G_z → slice selection (1D).
- G_y → phase encoding (2D).
- G_x → frequency encoding (2D).
- K-space is filled and then transformed (IFFT) to get image.



COMPARISON TABLE

Feature	Frequency Encoding (Readout)	Phase Encoding	Slice Selection
Gradient Axis	G_x	G_y	G_z
When Applied	During signal acquisition	Before readout (short blip)	During RF pulse
Effect	Changes frequency (ω)	Changes phase (ϕ)	Selects slice
Encodes	x-position	y-position	z-position
k-space Role	Fills k_x (horizontal)	Steps through k_y (vertical)	Selects 2D slice
Dimension	2D (line readout)	2D (line selection)	1D (slice)

✓ ADVANTAGES

- Accurate spatial localization.
- High spatial resolution achievable.
- Flexible control of resolution, FOV, and slice thickness.
- Works for 1D, 2D and 3D imaging.
- Basis for all MRI imaging techniques.

⚠ LIMITATIONS

- Longer scan time for fine resolution (more k-space lines).
- Sensitive to motion (phase errors).
- Gradient imperfections → artifacts.

★ HIGH-YIELD EXAM FACTS

- Frequency encoding → readout gradient (G_x).
- Phase encoding → phase gradient (G_y blip).
- Slice selection → slice selection gradient (G_z) with RF pulse.
- Resolution $\propto \frac{1}{D}$ matrix size in that direction.
- Bandwidth set by readout gradient (G_x).
- k-space center (low freq.) → image contrast.
- Edges of k-space (high freq.) → image detail.



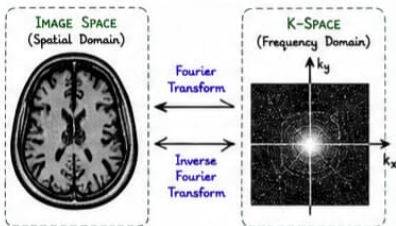
REMEMBER : SELECT the slice (G_z) → ENCODE the phase (G_y) → READ the frequency (G_x) → REPEAT to fill k-space → IFFT to build the image.

★ K-SPACE ★

① INTRODUCTION

- K-space (reciprocal space or Fourier space) is the space in which MR data are acquired.
- It is the raw data domain; image space is the reconstructed domain.
- Every point in k-space represents a specific combination of spatial frequencies.
- Center of k-space → low spatial frequencies (overall image contrast).
- Periphery of k-space → high spatial frequencies (image detail & edges).

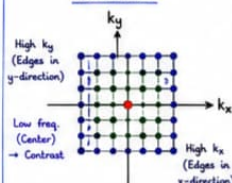
IMAGE SPACE VS K-SPACE



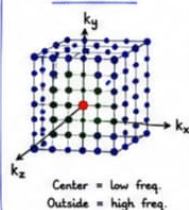
② WHAT IS K-SPACE?

- A 2D (k_x - k_y) or 3D (k_x - k_y - k_z) matrix that contains frequency information of the object.
- k_x (frequency encoding) → horizontal axis.
- k_y (phase encoding) → vertical axis.
- k_z (slice encoding) → through-plane (for 3D).
- Units: cycles per meter (m^{-1}).
- The signal at each (k_x , k_y) point is a complex number (amplitude + phase).

2D K-SPACE



3D K-SPACE

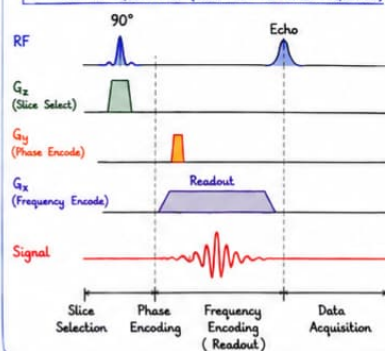


KEY POINTS

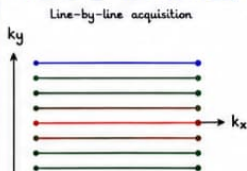
- More samples in k-space → better resolution and less aliasing.
- Symmetry: For real objects, k-space is conjugate symmetric.
- Image contrast is mainly determined by central k-space.
- Spatial resolution (Δx , Δy) is determined by k-space extent (FOV).

③ HOW ARE DATA ACQUIRED AND HOW ARE IMAGES CREATED FROM THESE DATA?

A. DATA ACQUISITION (EXAMPLE: 2D GRE SEQUENCE)

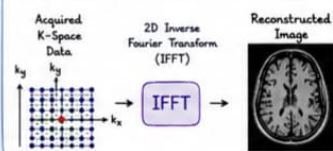


B. K-SPACE TRAVERSE (EXAMPLE)



- Each line = one phase encode (k_y)
- Readout along k_x
- Center lines acquired first in many sequences (better SNR)

C. FROM K-SPACE DATA TO IMAGE



MATHEMATICAL RELATIONSHIP

Image $f(x, y)$ is the inverse Fourier transform of k-space $F(k_x, k_y)$.

$$F(k_x, k_y) = \iint f(x, y) e^{-i2\pi(k_x x + k_y y)} dx dy \quad (\text{Forward FT})$$

$$f(x, y) = \iint F(k_x, k_y) e^{i2\pi(k_x x + k_y y)} dk_x dk_y \quad (\text{Inverse FT})$$

★ KEY POINTS (AT A GLANCE)

- Center of k-space → image contrast.
- Outer k-space → spatial resolution.
- Larger FOV → smaller k-space sampling density.
- Smaller FOV → larger k-space extent.
- Sampling only part of k-space → faster scan but may cause aliasing.
- Zero-filling increases matrix size → smoother image (not true resolution).



COMPARISON TABLE

Feature	Image Space	K-Space
Also called	Spatial domain	Reciprocal / Frequency domain
Represents	Object appearance	Spatial frequency information
Axes	x, y (and z)	k_x , k_y (and k_z)
Center	Not applicable	Low frequencies (contrast)
Periphery	Not applicable	High frequencies (details)
Data acquired?	No (after reconstruction)	Yes (raw data)
Transform	Result of IFFT	Obtained directly by MRI

✓ ADVANTAGES

- Efficient way to collect data.
- Enables trade-off between resolution, scan time and SNR.
- Basis for fast imaging techniques (EPI, spiral, radial, partial Fourier, parallel imaging).
- Allows manipulation (filtering, weighting) before image creation.

⚠ LIMITATIONS

- Incomplete k-space → artifacts (aliasing, blurring).
- Patient motion → phase errors → ghosting.
- Long scan for full k-space in conventional methods.
- Susceptible to system instabilities.

★ HIGH-YIELD EXAM FACTS

- K-space is in units of cycles/meter (m^{-1}).
- Image resolution: $\Delta x = 1/2 k_{x, \max}$, $\Delta y = 1/2 k_{y, \max}$
- K-space center determines image contrast.
- Phase errors in k-space → ghosting.
- Zero at k-space center = average signal intensity of the image.
- FFT/IFFT connect k-space ↔ image space



REMEMBER : CENTER → CONTRAST, OUTER → DETAILS, FULL K-SPACE → BEST IMAGE, PARTIAL K-SPACE → FASTER BUT RISK OF ARTIFACTS.

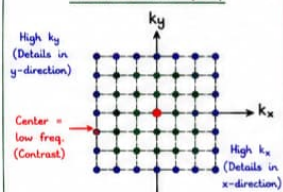


★ K-SPACE ★

① SOME IMPORTANT FACTS ABOUT K-SPACE

- K-space (reciprocal or Fourier space) is the raw data domain in MRI.
- It is a 2D (k_x - k_y) or 3D (k_x - k_y - k_z) grid of spatial frequencies.
- Units: cycles per meter (m^{-1}).
- Center of k-space (low k values) → low spatial frequencies → overall contrast.
- Periphery of k-space (high k values) → high spatial frequencies → fine details, edges.
- More samples in k-space → higher resolution & less aliasing.
- K-space is symmetric for real objects:
 $F(k_x, k_y) = F^*(-k_x, -k_y)$ (complex conjugate symmetry).
- Zero-filling increases matrix size → smoother image but NOT true increase in resolution.

K-SPACE MAP (2D)



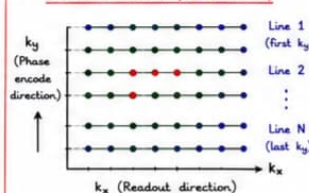
KEY

- Low freq. (center) → image contrast
- Mid freq. → structure
- High freq. (edges) → fine detail

② HOW DO PULSE SEQUENCES FILL K-SPACE?

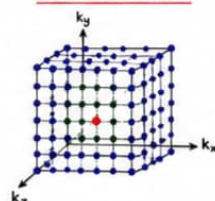
- MR signal is sampled at different (k_x, k_y [k_z]) positions.
- Phase encoding steps move along k_y (one line at a time).
- Frequency encoding reads out along k_x (entire line).
- Repetition for all phase encodes → fills entire k-space.

STEPWISE FILLING (2D EXAMPLE)



- Readout samples along k_x (one line)
- Each k_y step = one phase encode
- After N steps → full k-space

3D K-SPACE FILLING



- Center = low freq. (contrast)
- Outside = high freq. (detail)

CENTER-FIRST vs. OUTER-FIRST ACQUISITION

Center-First (typical)



- Acquire center early
- More robust to motion
- Preferred for most clinical imaging

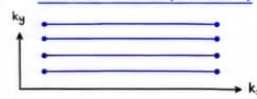
Outer-First



- Acquire periphery early
- Sensitive to motion
- Useful for special applications (e.g., MRA)

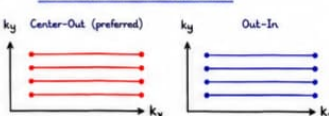
③ OPTIONS THAT FILL K-SPACE

A. LINE-BY-LINE (CARTESIAN)



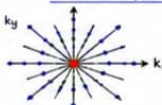
- Most common method.
- Straight lines; easy to implement.
- Susceptible to motion artifacts (ghosting).

B. CENTRIC / RE-ORDERING



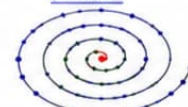
- Center-out → center of k-space acquired first.
- Better contrast stability in presence of motion.

C. RADIAL (SPIRAL-LIKE)



- Lines pass through center at many angles.
- Excellent motion robustness.
- Efficient for dynamic imaging.
- More complex reconstruction.

D. SPIRAL



- Continuous spiral trajectory.
- Very time efficient.
- Sensitive to off-resonance & system imperfections.
- Used in fast imaging (e.g., fMRI, diffusion).

★ KEY POINTS (AT A GLANCE)

- K-space center → image contrast.
- K-space periphery → image detail.
- Number of phase encodes (N_y) = number of lines.
- $FOV_y = 1 / \Delta k_y$ (Δk_y = phase encode step)
- $FOV_x = 1 / (N_x \cdot \Delta k_x)$ (Δk_x = readout dwell spacing)
- Bandwidth (BW) affects Δk_x :
 $\Delta k_x = BW / (\gamma \cdot N_x)$
- Larger BW → smaller Δk_x → less blurring & less chemical shift.



K-SPACE & IMAGE RELATIONSHIP

Low Frequencies (Center)



High Frequencies (Periphery)



Final Image



- Image = Inverse Fourier Transform of k-space data.
- Remove k-space lines → loss of corresponding detail.
- Truncation → ringing (Gibbs) artifacts.

✓ ADVANTAGES

- Understanding k-space helps optimize image quality.
- Enables trade-offs between resolution, time, and SNR.
- Basis for advanced techniques: parallel imaging, compressed sensing, partial Fourier, EPI.

⚠ LIMITATIONS

- Long scan for full k-space.
- Susceptible to motion (depends on sampling order).
- System imperfections cause artifacts.

★ HIGH-YIELD EXAM FACTS

- K-space units: cycles per meter (m^{-1}).
- Image resolution: $\Delta x = 1 / (2 k_{x \max})$, $\Delta y = 1 / (2 k_{y \max})$
- K-space center determines overall image contrast.
- Phase errors in k-space → ghosting in image.
- Zero at k-space center = average signal intensity.
- FFT (inverse Fourier transform) converts k-space → image.
- Partial k-space acquisition + symmetry → imaging possible.



REMEMBER : K-SPACE CENTER → CONTRAST | K-SPACE PERIPHERY → DETAIL | MORE K-SPACE SAMPLES → BETTER IMAGE QUALITY